

Atoms & Their Structure

≡ CUE ≡

- ★ What is an atom?
- ★ What are the subatomic particles?
- ★ Where are particles located?
- ★ What is the charge of each particle?
- ★ Why are atoms mostly empty space?

≡ NOTES ≡

An **atom** is the smallest unit of an element that still keeps the properties of that element.

Subatomic Particles:

- Proton (p^+) - positive charge
- Neutron (n^0) - no charge
- Electron (e^-) - negative charge

Where are particles located?

- Protons and neutrons are in the nucleus (center).
- Electrons move around the nucleus in the electron cloud (energy levels).

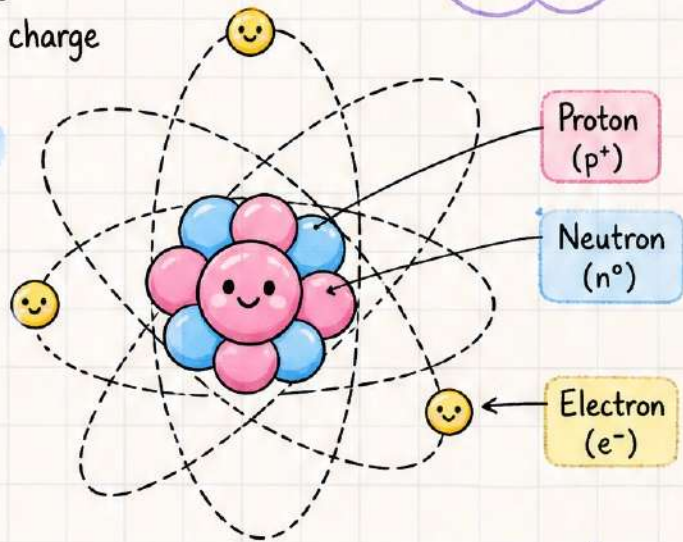
Charge of Each Particle

Particle	Symbol	Charge	Location
Proton	p^+	+1 (positive)	Nucleus
Neutron	n^0	0 (neutral)	Nucleus
Electron	e^-	-1 (negative)	Electron cloud

Why are atoms mostly empty space?

Because the nucleus is very small and the electrons are far away, most of the atom is empty space!

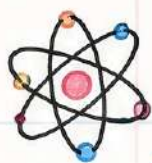
Atoms have a nucleus in the center with electrons around it.



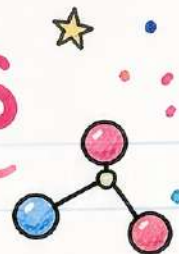
≡ SUMMARY ≡

- Atoms are the building blocks of matter.
- Atoms have protons (+), neutrons (0) in the nucleus.
- Electrons (-) move around the nucleus in the electron cloud.
- Atoms are mostly empty space.





Elements vs. Compounds

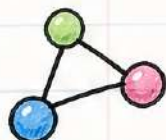


≡ CUE ≡

Difference between them?



Real-life examples?

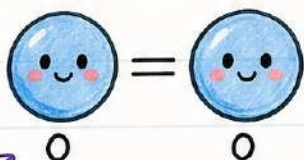


MAIN NOTES ≡

ELEMENTS	COMPOUNDS
Made of only one type of atom.	Made of two or more different types of atoms chemically bonded together.
Cannot be broken down into simpler substances by chemical means.	Can be broken down into simpler substances by chemical means.
Represented by a chemical symbol.	Represented by a chemical formula.
<u>Examples</u>: Au, O₂, Fe	<u>Examples</u>: H₂O, NaCl, CO₂

ELEMENT EXAMPLE:

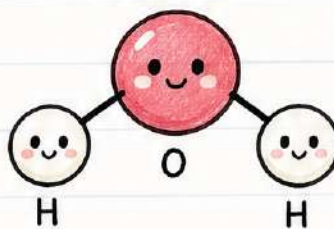
Oxygen (O₂)



Same atoms bonded together!

COMPOUND EXAMPLE:

Water (H₂O)



Different atoms bonded together!



Gold (Au)

A pure element.
Used in jewelry!



Salt (NaCl)

A compound made of sodium and chlorine.
Used in cooking!

≡ SUMMARY ≡

- Elements are made of **one type of atom** and cannot be broken down.
- Compounds are made of **two or more different atoms** bonded together and can be broken down.
- Both are the building blocks of matter and are all around us!





Practice Questions!



Try these!



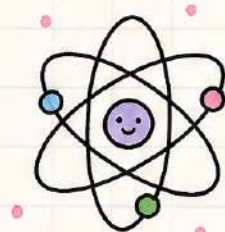
Check your knowledge



You've got this!



1. What are the three subatomic particles in an atom? For each one, tell its charge and where it is found in the atom.



2. An atom has 11 protons, 12 neutrons, and 11 electrons. What element is this? What is its atomic number?



3. If an atom has 17 electrons and 18 neutrons, what is the atomic number and mass number?



4. How do the number of protons in an atom determine what element it is?



5. Is H_2O an element or a compound? Explain how you know.



6. Name one element that is a gas at room temperature and one that is a solid. What are their chemical symbols?



Answer Key ✨

Cue

How did you do?



Great job!



Notes

1. Answer: C) Solid

A solid has a definite shape and volume. It keeps its shape no matter what.

2. Answer: B) Condensation

Condensation is when a gas (water vapor) cools down and turns into a liquid.

3. Answer: A) H_2O

H_2O is the chemical formula for water. It shows 2 hydrogen atoms and 1 oxygen atom.

4. Answer: D) All of the above

Mass, volume, and density are all physical properties. You can measure them without changing what the substance is.

5. Answer: B) Mixture

A mixture has two or more substances that are not chemically combined. You can often separate them by physical means.

6. Answer: C) 3 protons, 4 neutrons, 3 electrons

Atomic number 3 = 3 protons.

Mass number 7 = protons + neutrons $\rightarrow 7 - 3 =$ 4 neutrons.

For a neutral atom, electrons = protons $\rightarrow$ 3 electrons.

Learning takes practice,
and you're doing amazing! ❤️

